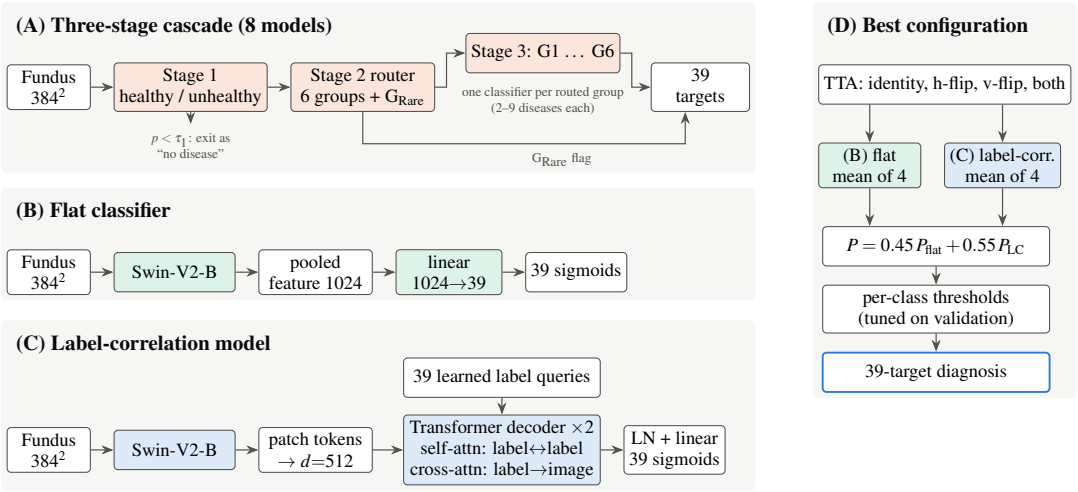

(A) Three-stage cascade (8 models)
Fundus 384²
Stage 1 healthy / unhealthy
$p < \tau_1$: exit as "no disease"
Stage 2 router 6 groups + G_Rare
Stage 3: G1 … G6
one classifier per routed group (2–9 diseases each)
39 targets
G_Rare flag
(B) Flat classifier
Fundus 384²
Swin-V2-B
pooled feature 1024
linear 1024→39
39 sigmoids
(C) Label-correlation model
39 learned label queries
Fundus 384²
Swin-V2-B
patch tokens → d=512
Transformer decoder ×2
self-attn: label↔label
cross-attn: label→image
LN + linear 39 sigmoids
(D) Best configuration
TTA: identity, h-flip, v-flip, both
(B) flat mean of 4
(C) label-corr. mean of 4
$P = 0.45\,P_{\text{flat}} + 0.55\,P_{\text{LC}}$
per-class thresholds (tuned on validation)
39-target diagnosis